

X-BAND DIGITAL MODULE

Complete Schematic Package

20 x 20 cm Space-Grade PCB

Direct RF Sampling Acquisition System

8-12 GHz X-Band | 500 MHz Bandwidth

ADC: ADC12DJ5200-SP (10.4 GSPS, 12-bit, Rad-Hard)

FPGA: XQRVC1902 (Rad-Hard, 1760-pin BGA)

Clock: LMX2615-SP + LMK04832-SP (45fs jitter)

Power: 6 rails (1.0V/1.8V/3.3V/2.5V), TPS7H5002-SP buck

Output: SpaceWire 2x 100 Mbps

Stackup: 12-layer Megtron6

Size: 200 x 200 mm

Revision 1.0 | 2026-09-10

SCHEMATIC SHEETS

01	Title Block	System overview and block diagram
02	RF Input	SMPA, TRF0208-SP balun, 50ohm termination
03	ADC12DJ5200-SP	12-bit 10.4GSPS ADC, JESD204C, power bypass
04	FPGA XQRVC1902	Rad-hard FPGA, JESD RX, SpaceWire, SPI, I2C
05	Clock Tree	LMX2615-SP, LMK04832-SP, VCXO 122.88MHz
06	Power Input	4.5V input, TVS, ferrite, bulk caps
07	Buck 1.0V Core	TPS7H5002-SP, 44A, 100nH, all passives
08	Buck 1.8V Aux	ISL70003ASEH, 6A, 2.2uH, all passives
09	Buck 3.3V I/O	ISL70003ASEH, 3A, 4.7uH, all passives
10	LDO 1.0V Analog	TPS7H1111-SP, AVDD for ADC
11	LDO 1.8V Digital	TPS7H1111-SP, DVDD
12	LDO 2.5V Clock	TPS7H1121-SP, VCLK

X-Band Digital Module - Sheet01 Title

File: Sheet01_Title.kicad_sch



X-Band Digital Module - Sheet02 RF Input

File: Sheet02_RF_Input.kicad_sch



X-Band Digital Module - Sheet03 ADC

File: Sheet03_ADC.kicad_sch



X-Band Digital Module - Sheet04 FPGA

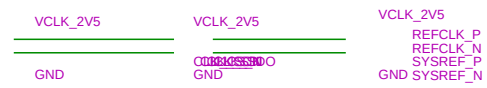
File: Sheet04_FPGA.kicad_sch

Schematic sheet (open in KiCad for full rendering)

Components: 0, Wires: 0

X-Band Digital Module - Sheet05 Clock

File: Sheet05_Clock.kicad_sch



X-Band Digital Module - Sheet06 PowerInput

File: Sheet06_PowerInput.kicad_sch



X-Band Digital Module - Sheet07 Buck 1V0

File: Sheet07_Buck_1V0.kicad_sch

Schematic sheet (open in KiCad for full rendering)

Components: 0, Wires: 0

X-Band Digital Module - Sheet08 Buck 1V8

File: Sheet08_Buck_1V8.kicad_sch

Schematic sheet (open in KiCad for full rendering)

Components: 0, Wires: 0

X-Band Digital Module - Sheet09 Buck 3V3

File: Sheet09_Buck_3V3.kicad_sch

Schematic sheet (open in KiCad for full rendering)

Components: 0, Wires: 0

X-Band Digital Module - Sheet10 LDO 1V0

File: Sheet10_LDO_1V0.kicad_sch

Schematic sheet (open in KiCad for full rendering)

Components: 0, Wires: 0

X-Band Digital Module - Sheet11 LDO 1V8

File: Sheet11_LDO_1V8.kicad_sch

Schematic sheet (open in KiCad for full rendering)

Components: 0, Wires: 0

X-Band Digital Module - Sheet12 LDO 2V5

File: Sheet12_LDO_2V5.kicad_sch

Schematic sheet (open in KiCad for full rendering)

Components: 0, Wires: 0

X-Band Digital Module - Sheet13 Sequencer

File: Sheet13_Sequencer.kicad_sch

Schematic sheet (open in KiCad for full rendering)

Components: 0, Wires: 0

X-Band Digital Module - Sheet14 Connectors

File: Sheet14_Connectors.kicad_sch

Schematic sheet (open in KiCad for full rendering)

Components: 0, Wires: 0

X-Band Digital Module - Sheet15 TestPoints

File: Sheet15_TestPoints.kicad_sch

Schematic sheet (open in KiCad for full rendering)

Components: 0, Wires: 0

X-Band Digital Module - Sheet16 Decoupling

File: Sheet16_Decoupling.kicad_sch

Schematic sheet (open in KiCad for full rendering)

Components: 0, Wires: 0

X-Band Digital Module - Sheet17 PowerTree

File: Sheet17_PowerTree.kicad_sch

Schematic sheet (open in KiCad for full rendering)

Components: 0, Wires: 0

X-Band Digital Module - Main Schematic - Hierarchical Sheets

File: XBandDigitalModule.kicad_sch

Schematic sheet (open in KiCad for full rendering)

Components: 0, Wires: 0